

Unit 13, Lesson 4a: Weighted Averages of Directed Line Segments



Benchmark

MA.912.GR.3.1 Determine the weighted average of two or more points on a line.



Learning Target

- ☐ I can determine the weighted average of two or more points on a directed line segment.
- ☐ I can explain how changing the weight moves the weighted average of a point on a directed line segment.



13.4a.1 Warm-Up: Exploring Rays

Four rays are shown.

- What do you notice about the four rays?

Each ray includes a line segment. These line segments are known as *directed line segments*. When naming a directed line segment, the first letter is the endpoint, and the second letter indicates the direction.

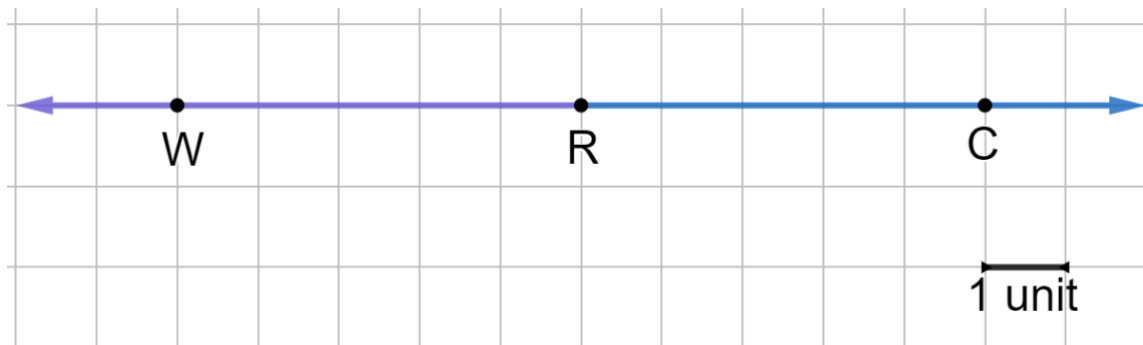
- Which rays include the directed line segment HD ?
- Which rays include the directed line segment DH ?

Ray 1	
Ray 2	
Ray 3	
Ray 4	



13.4a.2 Collaboration: Parts of Directed Line Segment

Rays RW and RC are shown on a grid, where each grid line represents 1 unit.



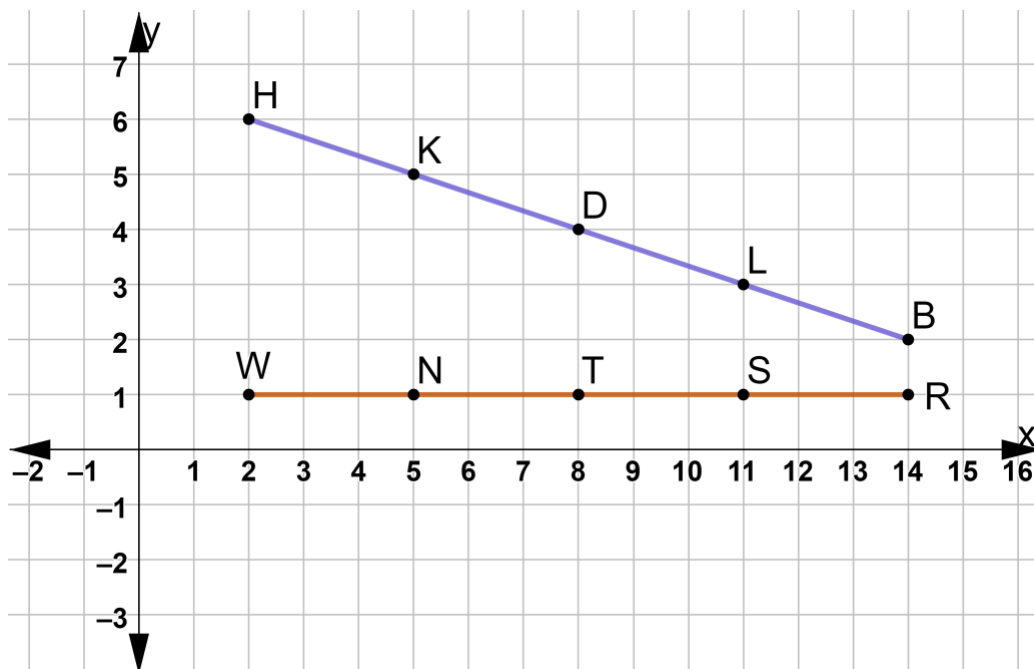
1. Complete the table for each

Ray	Directed Line Segment	Direction (Left or Right)	Length of the Directed Line Segment
$\overrightarrow{RW}$			
$\overrightarrow{WC}$			
$\overrightarrow{RC}$			
$\overrightarrow{CW}$			

2. What is the ratio of the length of the directed line segment RW to the length of directed line segment RC ?
3. Complete the statement.

Point R is _____ between point W and point C .

Directed line segment HB and directed line segment WR are shown on the coordinate grid.



4. Complete each statement.
 - a. Directed line segment HB is divided into _____ parts.
 - b. Directed line segment WR is divided into _____ parts.
5. Point N divides directed line segment WR into the ratio of $\frac{WN}{NR}$. Point K divides directed line segment HB into the ratio of $\frac{KH}{BK}$. Discuss with your partner why these ratios have the same value. Summarize your discussion.

6. Complete the statement.

The ratios $\frac{WN}{NR}$ and $\frac{KH}{BK}$ are

- ☐ part : part
☐ part : whole
- ratios.

7. Point K is closer to point H , so the ratio 1:3 is known as a *weighted average*. When a point does not divide a line segment at its midpoint, the result is that the line segment is divided so that one of the endpoints has more weight.
- a. On directed line segment HB , which endpoint is closer to point K ?

Point K divides directed line segment HB so that $HK : KB$ is 1 : 3. The formula $\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } B)$ can be used to determine the location of point K .

- b. Discuss with your partner why the ratio is 1 : 3 but the formula uses a denominator of 4. Summarize your discussion.
8. Use the x -values to determine $\frac{3}{4}(\text{point } H) + \frac{1}{4}(\text{point } B)$.
9. Use the x -values to determine $\frac{3}{4}(\text{point } W) + \frac{1}{4}(\text{point } R)$.
10. What do you notice about the results of the two sums?
11. Point S divides directed line segment WR so that $WS : SR$ is 3 : 1. Complete the fractions to write the formula for the weighted average that would partition directed line segment WR resulting in point S .

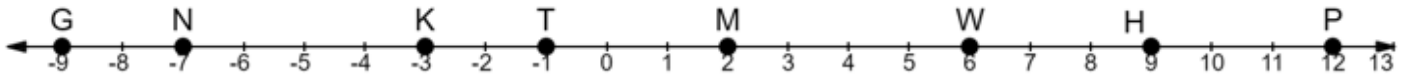
$$\frac{\boxed{}}{\boxed{}}(\text{point } W) + \frac{\boxed{}}{\boxed{}}(\text{point } R)$$

12. Verify your formula with your partner and make any necessary changes. Then, use your formula to show that it results in the x -value of point S .



13.4a.3 Exploration: Weighted Averages on a Number Line

A number line is shown with several points plotted.



1. Complete the statements.
 - a. Directed line segment WG has a length of _____ units.
 - b. Directed line segment KP has a length of _____ units.
 - c. The ratio $\frac{WK}{WG}$ is _____.
 - d. The ratio $\frac{KW}{KP}$ is _____.
2. Discuss with your partner whether point K on directed line segment WG has the same weighted average as point W on directed line segment KP . Summarize your discussion.

The ratio of $WK : KG$ can be written as $9 : 6$ or $3 : 2$. This is the weighted average. If you count 9 spaces from point W in the direction of point G , the result is point K . And if you go 6 spaces from point K in the direction of point G is point G .

Although it is easy to count in this situation it is not always easy to count spaces because a point can be **any** real number on a number line.

3. Complete the statements.
 - a. The ratio $\frac{WK}{WG}$ is a

☐ part : part
☐ part : whole

 ratio.
 - b. The ratio $\frac{WK}{KG}$ is a

☐ part : part
☐ part : whole

 ratio.

Point W partitions directed line segment KP so that point P weighs more than point K . This means that point W is closer to point P because point P is “heavier” than point K .

- What is the part:whole ratio that is the result of point W partitioning directed line segment KP ?
- What is the part:part ratio that is the result of point W partitioning directed line segment KP ?
- Discuss with your partner how the part:whole ratio and the part:part ratio could be used to determine the formula that would give the location of Point W on the number line. Then, complete the formula.

$$\frac{\boxed{}}{\boxed{}}(\text{point } K) + \frac{\boxed{}}{\boxed{}}(\text{point } P)$$

- Complete the table by determining the endpoint of each directed line segment that would be “heavier” given each point that partitions the segment.

Directed Line Segment	Point That Partitions the Segment	Endpoint That Is Heaviest
$\overrightarrow{GM}$	T	
$\overrightarrow{MG}$	T	
$\overrightarrow{TH}$	W	
$\overrightarrow{PN}$	M	
$\overrightarrow{PW}$	H	

8. Point T partitions directed line segment GM .

Directed Line Segment	Point That Partitions the Segment	Ratio	Numerical Ratio	Formula
$\overline{GM}$	T	$GT : TM$	$8 : 3$	$\frac{3}{11}(\text{point G}) + \frac{8}{11}(\text{point M})$

- a. Discuss with your partner how the formula in the table relates to the numerical ratio and the heaviest endpoint. Summarize your discussion.

- b. Use the values of point G and point M in the formula $\frac{3}{11}(\text{point G}) + \frac{8}{11}(\text{point M})$ to verify that the result is point T .

9. Complete the table.

Directed Line Segment	Point That Partitions the Segment	Ratio	Numerical Ratio	Formula
$\overline{MG}$	T	$MT : TG$		
$\overline{TH}$	W			
$\overline{PN}$	M			
$\overline{PW}$	H			



13.4a.4 Guided Instruction: Weighted Averages on a Number Line

1. A number line is shown with several points labeled.



- Determine the ratio of $HG : GB$.
- For the directed line segment HB , which endpoint is heaviest in the ratio $HG : GB$?
- Write the formula for the directed line segment HB that would result in point G .
- Which point on directed line segment HB and directed line segment BH would have the same formula?
- Which point on directed line segment GR partitions the line segment into line segments in a ratio of 4 : 1?
- For the directed line segment GR , which endpoint is heaviest in the ratio 2 : 3?
- Write the formula for the directed line segment GR that would partition the line segment in a ratio of 7 : 3.

2. Points H , N and W are on a number line. Point H is located at 7. Point W is located at -2 . Point W partitions directed line segment NH in the ratio $1 : 3$.
- Draw a number line that shows points H and W are on a number line.
 - Use the number line in part a to show how point N could be determined.
 - For the ratio $1 : 3$, which endpoint of the directed line segment NH is the heaviest?
 - Use the formula for partitioning line segments to write an equation that can be used to determine the location of endpoint N of directed line segment NH .
 - Use the equation to determine the location of endpoint N of directed line segment NH .
3. Point K partitions directed line segment RD in the ratio $3 : 5$. On a number line, point R is located at -23 and point K is located at 1. Determine the location of point D .
4. Point Y partitions directed line segment HT in the ratio $1 : 1$. On a number line, point T is located at $-4\frac{3}{4}$ and point Y is located at $-8\frac{1}{8}$. Determine the location of point H .

5. Point N partitions directed line segment KW in the ratio $9 : 16$. On a number line, point N is located at $-8\frac{2}{3}$ and point W is located at $55\frac{1}{3}$. Determine the location of point K .



A midpoint calculation weighs points on a line segment equally. In a **weighted average** calculation, points on a line segment can be weighted differently.

6. Discuss with your partner how to determine from a ratio if a point partitions a directed line segment equally. Summarize your discussion.



13.1a.5 Guided Practice: Weighted Averages

1. Point B partitions directed line segment YN in the ratio $7 : 13$. On a number line, point N is located at 17.4 and point B is located at 38.4 . Determine the location of point Y .
2. Point W partitions directed line segment NB . On a number line, point B is located at $-22\frac{3}{4}$, point N is located at $8\frac{1}{4}$, and point W is located at $-\frac{7}{4}$. Determine the weighted average.



13.4a.6 Your Turn!

1. Point L partitions directed line segment TS . On a number line, point S is located at -72 , point L is located at -32 , and point T is located at 43 . Determine the weighted average.

2. Point R partitions directed line segment TK in the ratio $1 : 1$. On a number line, point T is located at $-\frac{49}{2}$ and point R is located at $-\frac{45}{8}$. Determine the location of point K .



13.4a.7 Wrap-Up: Ticket Out the Door

1. Point B partitions directed line segment LF in the ratio $5 : 7$. On a number line, point F is located at 13.62 and point B is located at -3.88 . Determine the location of point L .